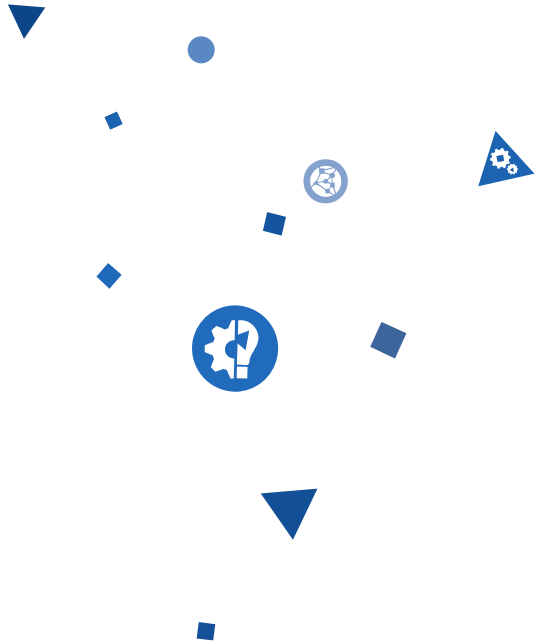


Java performance techniques. The cost of HotSpot runtime optimizations.

Ionuţ Baloşin

Software Architect
ibalosin@luxoft.com
@ibalosin

VOXXED DAYS VIENNA

About Me

Ionuţ Baloşin

Software Architect @ ww.luxoft.com

- **10+** years of software development experience

Technical Trainer @ www.luxoft-training.com

- Java Performance and Tuning
- Software Architecture

Conference Speaker

Agenda

- Just In Time Compiler Introduction
- Uncommon Traps
- Null Sanity Checks
- Devirtualization
- Loop Unrolling
- String Deduplication
- Biased Locking
- Inlining & Concurrency Implications
- Methods that do not JIT

Take Away

Better understanding about what happens under the hood inside HotSpot JVM from JIT standpoint.

Learn about few JIT optimizations.

Cases when few JIT optimizations are prevented. How can be tackled such situations.

In a nutshell: it helps you to better tune your application from performance standpoint.

My Configuration

Hardware

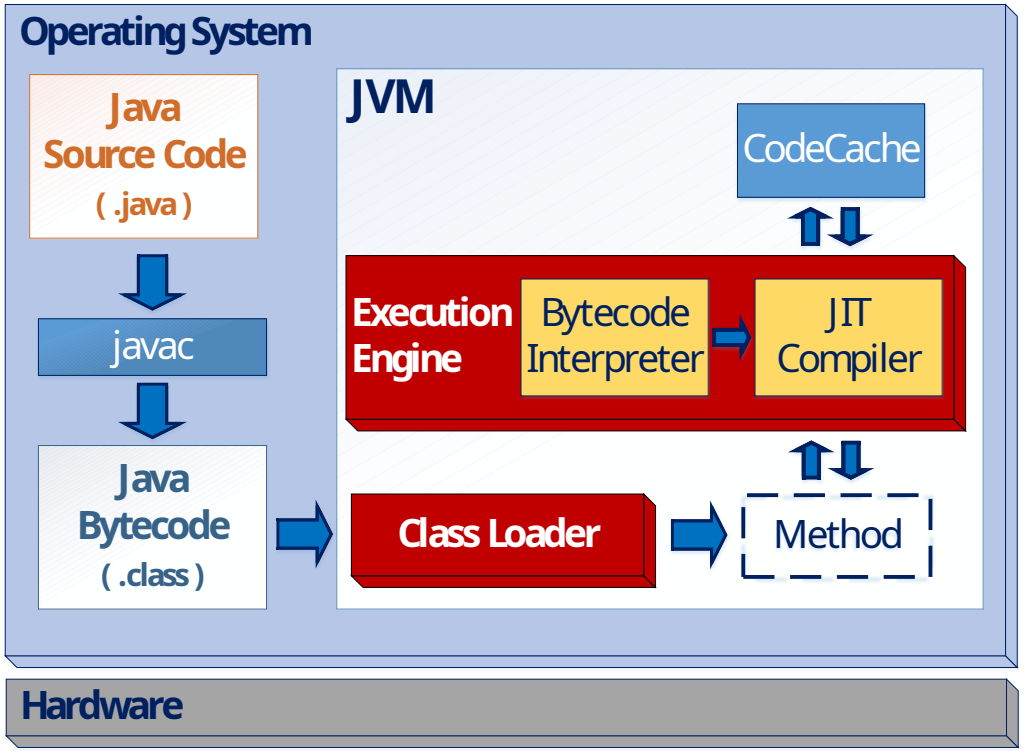
- Intel i7-6700HQ Skylake
- 16GB DDR4 2133 MHz
- 1TB 5400 RPM+512GB PCIe SSD

Software

- Ubuntu 16.04.2 & Win 10 Enterprise x64
- JDK 1.8.0_121 x64
- JITWatch
- HPjmeter

Just In Time Compiler Introduction

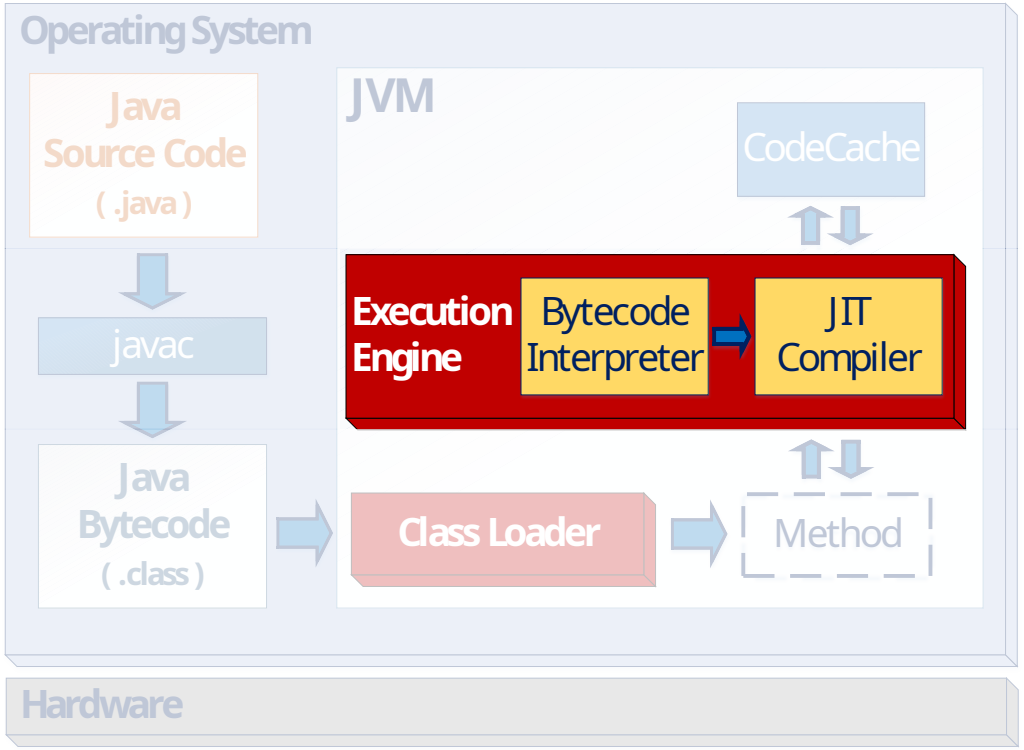
Just In Time Compiler - HotSpot



Just In Time Compiler - HotSpot

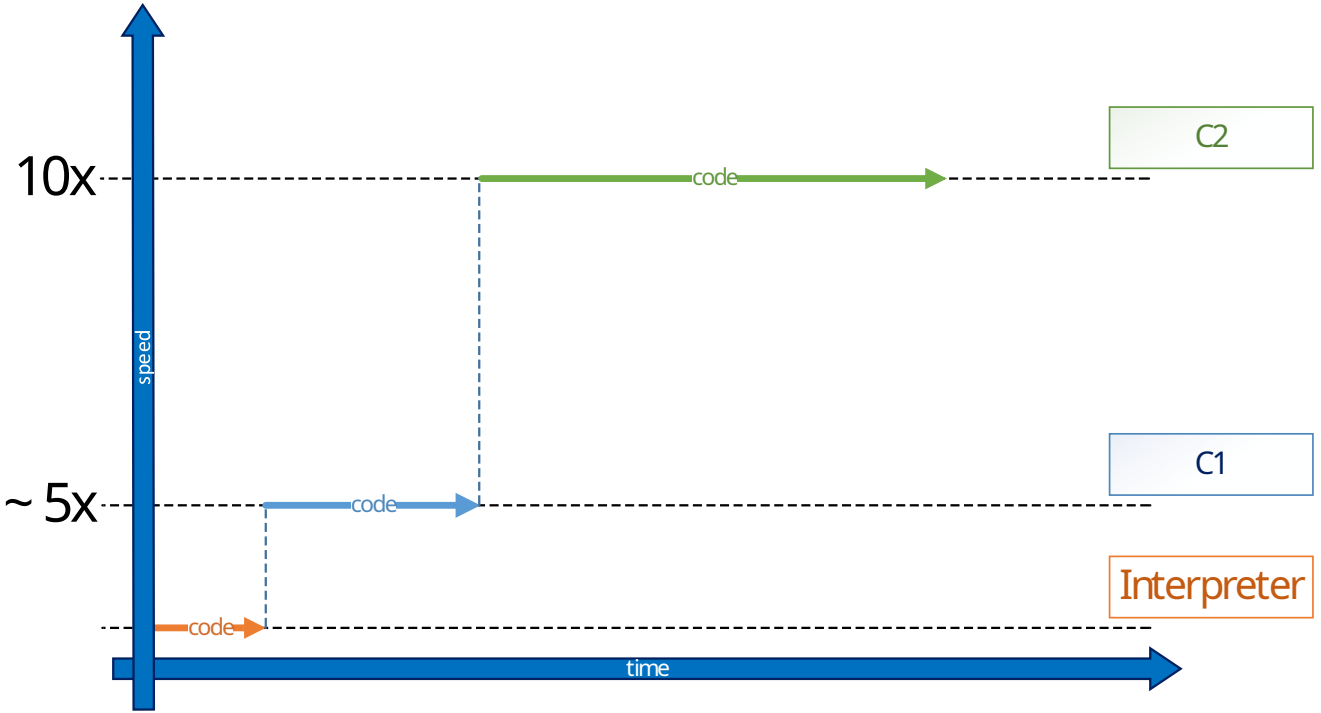
JIT Compiler

- Client (C1)
- Server (C2)
- Tiered Mode (C1 + C2)
  - Tier 0 = Interpreter
  - Tier 1 = C1 w/o profiling
  - Tier 2 = C1 w/ basic profiling
  - Tier 3 = C1 w/ full profiling
  - Tier 4 = C2



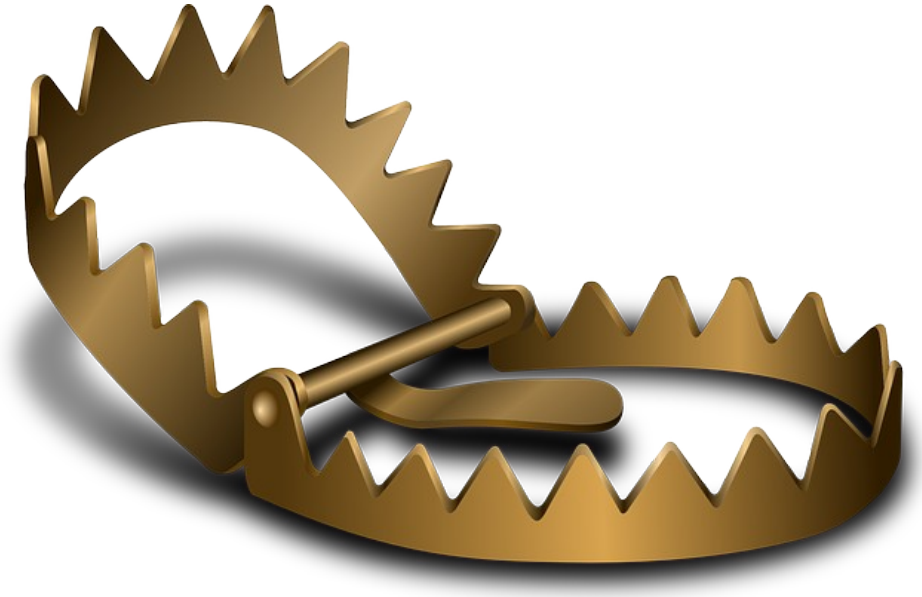
Just In Time Compiler - HotSpot

Tiered Compilation Flows



Uncommon Traps

Uncommon Traps



Uncommon trap is nothing more than a trampoline back to the Interpreter.
The compiled code is de-optimized and is flagged as being unusable.

Uncommon Traps

- **Code Sample**

```
int sum = 0;
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(true);
}
```

```
private int hotMethod(boolean condition) {
    int counter;

    if (condition == true)
        counter = 1;
    else
        counter = 2;

    return counter;
}
```

Uncommon Traps

$> java -XX:+PrintAssembly -XX:PrintAssemblyOptions=intel -XX:-TieredCompilation UncommonTraps.class

```
# {method} {0x00000000104004f0} 'hotMethod'
# parm0:    rdx       = boolean
#           [sp+0x20]  (sp of caller)
[Entry Point]
...
cmp edx,0x1    ; if (condition == true)
jne L0000      ;*if_icmpne
               ; - UncommonTraps::hotMethod@2 (line 52)
mov eax,0x1    ; returns 1
...
```

Uncommon Traps

$> java -XX:+PrintAssembly -XX:PrintAssemblyOptions=intel -XX:-TieredCompilation UncommonTraps.class

```
# {method} {0x00000000104004f0} 'hotMethod'
# parm0:    rdx       = boolean
#           [sp+0x20]  (sp of caller)
[Entry Point]
...
cmp edx,0x1    ; if (condition == true)
jne L0000      ;*if_icmpne
               ; - UncommonTraps::hotMethod@2 (line 52)
mov eax,0x1    ; returns 1
...
```

```
L0000: mov ebp,edx
       mov edx,0xffffff65
       xchg ax,ax
       call 0x00000000034157a0  ; OopMap{off=48}
                                ;*if_icmpne
                                ; - UncommonTraps:hotMethod@2 (line 52)
                                ;   {runtime_call}
```

uncommon trap

JIT optimized *(condition == true)* branch. To handle *(condition == false)* an uncommon trap was added

Uncommon Traps

```
int sum = 0;
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(true);
}
```

optimized & compiled

- Let's check now how long takes one hotMethod(true) call

```
hotMethod(true);      // ~ 300 ns
```

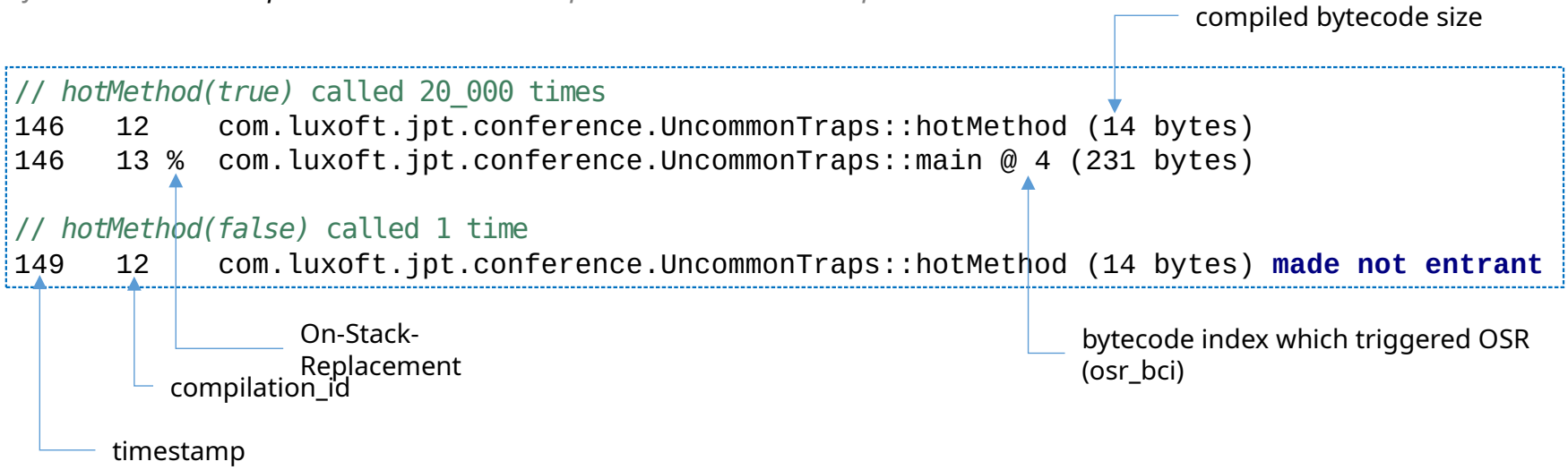
- But … what if calling hotMethod(false) immediately after!?

```
hotMethod(false);      // ~ 30,000 ns      Ouch ☹ !
```

Why so big difference between these two calls !?

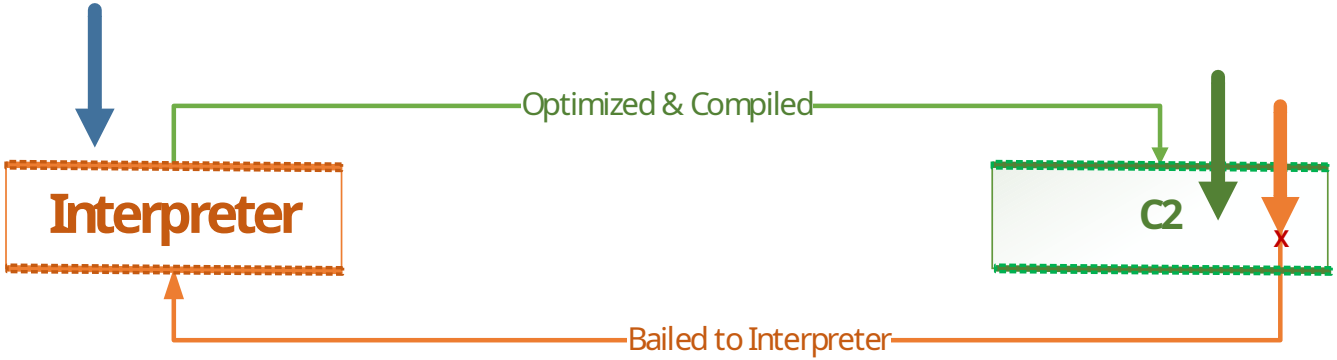
Uncommon Traps

$> java -XX:+PrintCompilation -XX:-TieredCompilation UncommonTraps.class



**** made not entrant** - no future callers to the broken code!*

Uncommon Traps



Thread hitting the trap

Thread without trap

Thread calling the same method after trap was hit by other Thread

*Current thread which hits the trap bails back to **Interpreter**.*

Another thread running the same method do not bail to Interpreter as long as it does not hit the trap.

*After bailing to **Interpreter** all new method calls by other threads will use this method version.*

Uncommon Traps

- Let's try to optimize hotMethod(boolean) for both *true* and *false* conditional branches

```
int sum = 0;
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(true);
}
// ...
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(false);
}
```

$> java -XX:+PrintCompilation -XX:-TieredCompilation UncommonTraps.class

```
// hotMethod(true) called 20_000 times
162   12    com.luxoft.jpt.conference.UncommonTraps::hotMethod (14 bytes)
162   13 %  com.luxoft.jpt.conference.UncommonTraps::main @ 4 (74 bytes)

// hotMethod(false) called 20_000 times
163   12    com.luxoft.jpt.conference.UncommonTraps::hotMethod (14 bytes) made not entrant
163   14    com.luxoft.jpt.conference.UncommonTraps::hotMethod (14 bytes)
163   15 %  com.luxoft.jpt.conference.UncommonTraps::main @ 28 (74 bytes)
```

Uncommon Traps

$> java -XX:+PrintAssembly -XX:PrintAssemblyOptions=intel -XX:-TieredCompilation UncommonTraps.class

```
# {method} {0x000000000fa906c0} 'hotMethod' '(Z)I' in 'UncommonTraps'
# parm0:    rdx       = boolean
#           [sp+0x20]  (sp of caller)
[Entry Point]
...
mov r11d,0x2
mov eax,0x1
cmp edx,0x1      ; if (condition == true)
cmovne eax,r11d  ;*iload_1
                 ; - UncommonTraps::hotMethod@12 (line 59)
...
```

Not any uncommon trap, JIT falls back to a classic way to handle branches!

Uncommon Traps

Sum Up

If a conditional statement was following only a single branch, an uncommon trap is a good idea to optimize this.

In case the other branch was called, JIT falls back to the classic way of handling conditional statements (e.g. cmovne).

Triggering uncommon trap is costly since they fall back in Interpreter.

Uncommon Traps

World of Uncommon Traps

Null Sanity Checks

Null Sanity Checks

- **Code Sample**

```
int sum = 0;
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(17);
}
```

```
private int hotMethod(Integer anInt) {
    int counter = 0;

    if (null != anInt)
        counter = anInt * 2;

    return counter;
}
```

Null Sanity Checks

$> java -XX:+PrintAssembly -XX:PrintAssemblyOptions=intel -XX:-TieredCompilation NullSanityChecks.class

```
# {method} {0x000000000ffb0668} 'hotMethod' '(Ljava/lang/Integer;)I' in
'NullSanityChecks'
# parm0:    rdx:rdx   = 'java/lang/Integer'
#           [sp+0x30]  (sp of caller)
[Entry Point]
...
mov eax,DWORD PTR [rdx+0xc]  ; implicit exception: dispatches to 0x0000000002f7f1dd
shl eax,1  ;*imul
           ; - NullSanityChecks::hotMethod@12 (line 33)
...
```

Null Sanity Checks

$> java -XX:+PrintAssembly -XX:PrintAssemblyOptions=intel -XX:-TieredCompilation NullSanityChecks.class

```
# {method} {0x000000000ffb0668} 'hotMethod' '(Ljava/lang/Integer;)I' in
'NullSanityChecks'
# parm0:    rdx:rdx   = 'java/lang/Integer'
#           [sp+0x30]  (sp of caller)
[Entry Point]
...
mov eax,DWORD PTR [rdx+0xc]  ; implicit exception: dispatches to 0x0000000002f7f1dd
shl eax,1  ;*imul
           ; - NullSanityChecks::hotMethod@12 (line 33)
...
```

implicit NULL check

```
0x0000000002f7f1dd: mov QWORD PTR [rsp],rdx
                    mov edx,0xffffff65
                    nop
                    call 0x0000000002f557a0  ; OopMap{[0]=Oop off=44}
                                             ;*if_acmpeq
                                             ; - NullSanityChecks::testNullChecks@4 (line 32)
                                             ;   {runtime_call}
```

NULL handler
- uncommon trap -

JIT leverages on default OS signals (e.g. SEG_FAULT) to throw NPEs

Null Sanity Checks

- Let’s try to optimize hotMethod(Integer) for both *NULL* and *!NULL* branches

```
int sum = 0;
for (int i = 0; i < 20_000; i++)
    sum += hotMethod(17);

sum += hotMethod(null);

for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(17);
}
```

Null Sanity Checks

```
# {method} {0x000000000fa60658} 'hotMethod' '(Ljava/lang/Integer;)I' in 'NullSanityChecks'
# parm0:    rdx:rdx   = 'java/lang/Integer'
#           [sp+0x20]  (sp of caller)
[Entry Point]
...
        test rdx,rdx
        je L0001                        ;*if_acmpeq
                                        ; - NullSanityChecks::hotMethod@4 (line 31)
        mov eax,DWORD PTR [rdx+0xc]
        shl eax,1                       ;*iload_1
                                        ; - NullSanityChecks::hotMethod@14 (line 34)
L0000:  add rsp,0x10
        pop rbp
...
        ret
L0001:  xor eax,eax
        jmp L0000
...
```

Not any uncommon trap, JIT falls back to a classic way to handle NULL!

Null Sanity Checks

Sum Up

NULL checks relying on signals (e.g. SEG_FAULT) are natively provided by hardware and handled by OS.

In case NULL branch is triggered JIT falls back to the classic way of handling NULL conditional statements (which is also costly).

Do not let NULL happening a lot in your application becomes it slows it down.

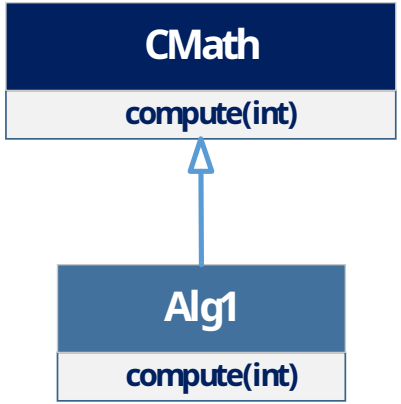
Devirtualization

Devirtualization

- **Code Sample**

```
double sum = 0;
CMath alg1 = new Alg1();
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(alg1, i);
}
```

```
private double hotMethod(CMath math, int i) {
    return math.compute(i);    // return (double) (const * i)
}
```



Devirtualization

$> java -XX:+PrintAssembly -XX:PrintAssemblyOptions=intel -XX:-TieredCompilation CHAVirtualCall.class

```
# {method} {0x000000000ffa05a0} 'hotMethod' '(LCMath;I)D' in 'CHAVirtualCall'
# parm0:    rdx:rdx   = 'CMath'
# parm1:    r8        = int
#           [sp+0x20]  (sp of caller)
[Entry Point]
...
vcvtsi2sd xmm0,xmm0,r8d                                ; convert r8:int into xmm0:double precission
vmulsd xmm0,xmm0,QWORD PTR [rip+0xfffffffffffffc2] # 0x000000000302eec0
                                                       ; multiply xmm0 by const
                                                       ;*dmul
                                                       ; - Alg1::compute@5 (line 58)
                                                       ; - CHAVirtualCall::hotMethod@2 (line 37)
                                                       ;   {section_word}
...
```

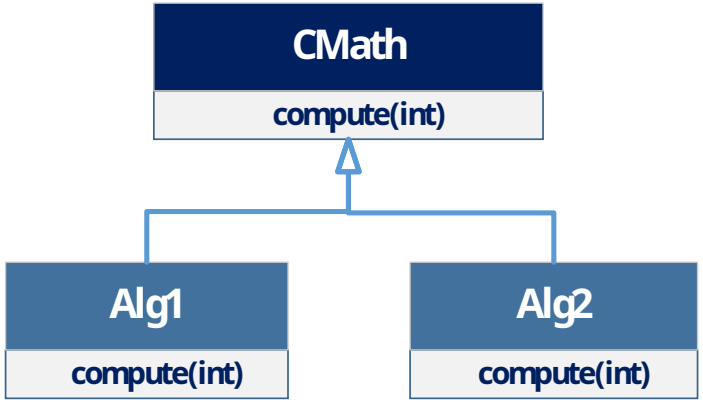
Monomorphic call (single target) is inlined without any uncommon trap.

Devirtualization

- **Code Sample**

```
double sum = 0;
CMath alg1 = new Alg1();
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(alg1, i);
}

CMath alg2 = new Alg2();
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(alg2, i);
}
```



Devirtualization

$> java -XX:+PrintAssembly -XX:PrintAssemblyOptions=intel -XX:-TieredCompilation CHAVirtualCall.class

```
# {method} {0x000000000fde0610} 'hotMethod' '(LCMath;I)D' in 'CHAVirtualCall'
# parm0:    rdx:rdx   = 'CMath'
# parm1:    r8        = int
#           [sp+0x30]  (sp of caller)
[Entry Point]
...
       vcvtsi2sd xmm0,xmm0,r8d  ;*i2d  ; - Alg2::compute@1 (line 69)
       cmp r10d,0xf800c082              ;   {metadata('Alg1')}
       je L0000
       cmp r10d,0xf800c0c1              ;   {metadata('Alg2')}
       jne L0002
       vmulsd xmm0,xmm0,QWORD PTR [rip+0xfffffffffffffb9]  # 0x0000000002e2b2a8
                                                           ; - Alg2::compute@5 (line 69)
       jmp L0001                                           ; method return
L0000: vmulsd xmm0,xmm0,QWORD PTR [rip+0xfffffffffffffa7]  # 0x0000000002e2b2a0
                                                           ; - Alg1::compute@7 (line 58)
...

L0002: call 0x0000000002df57a0  ; OopMap{rbp=Oop off=88}
                                ; - CHAVirtualCall::hotMethod@2 (line 37)
...
```

uncommon trap

Bimorphic call (two targets) is also inlined but uncommon trap is added.

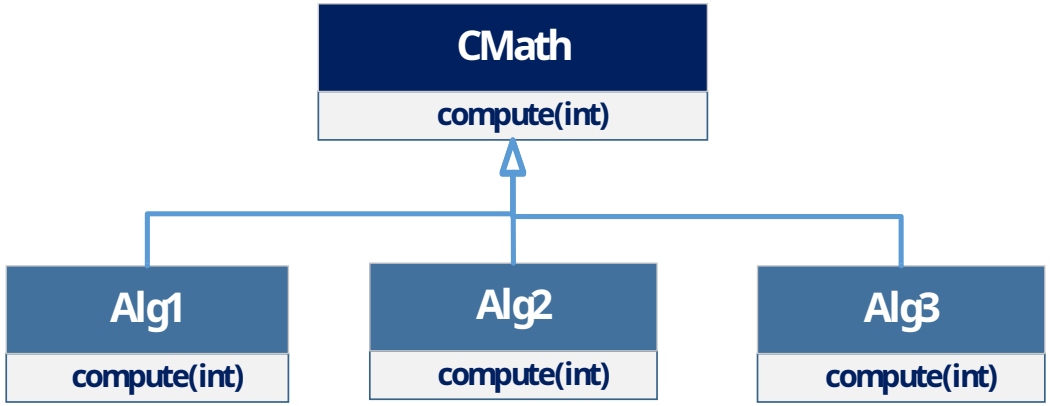
Devirtualization

- **Code Sample**

```
double sum = 0;
CMath alg1 = new Alg1();
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(alg1, i);
}

CMath alg2 = new Alg2();
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(alg2, i);
}

CMath alg3 = new Alg3();
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(alg3, i);
}
```



Devirtualization

$> java -XX:+PrintAssembly -XX:PrintAssemblyOptions=intel -XX:-TieredCompilation CHAVirtualCall.class

```
# {method} {0x0000000010260678} 'hotMethod' '(LCMath;I)D' in 'CHAVirtualCall'
# parm0:    rdx:rdx   = 'CMath'
# parm1:    r8        = int
#           [sp+0x20]  (sp of caller)
[Entry Point]
...
movabs rax,0xffffffffffffffff
call 0x00000000032063e0  ; OopMap{off=28}
                         ;*invokevirtual compute
                         ; - CHAVirtualCall::hotMethod@2 (line 37)
                         ;   {virtual_call}
...
```

Megamorphic call is not anymore inlined, neither any uncommon trap is added.

Devirtualization

Sum Up

Monomorphic calls are inlined without any added uncommon trap.

Bimorphic calls are inlined but pay the cost of jump instructions and uncommon trap.

Megamorhic calls are handled as virtual calls.

The cost from monomorphic to megamorphic calls is ~ 3-7 ns.

Loop Unrolling

Loop Unrolling

- **Code Sample**

```
int sum = 0;
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod(array);
}
```

```
private int hotMethod(int[] array) {
    int sum = 0;

    for (int i = 0; i < array.length; i++) {
        sum += array[i];
    }

    return sum;
}
```

Loop Unrolling

$> java -XX:+PrintAssembly -XX:PrintAssemblyOptions=intel -XX:-TieredCompilation LoopUnrolling.class

```
# {method} {0x00000000103f0538} 'hotMethod' '([I)I' in 'LoopUnrolling'
# parm0:    rdx:rdx   = '[I'
#           [sp+0x20]  (sp of caller)
[Entry Point]
...
       mov eax,DWORD PTR [rdx+0x10]  ;*iaload – peels the first loop iteration
                                     ; - LoopUnrolling::hotMethod@13 (line 25)
...
       mov r9d,0x1
...
L0000: add eax,DWORD PTR [rdx+r9*4+0x10]
       movsxd r10,r9d
       add eax,DWORD PTR [rdx+r10*4+0x14]
       add eax,DWORD PTR [rdx+r10*4+0x18]
       add eax,DWORD PTR [rdx+r10*4+0x1c]
       add eax,DWORD PTR [rdx+r10*4+0x20]
       add eax,DWORD PTR [rdx+r10*4+0x24]
       add eax,DWORD PTR [rdx+r10*4+0x28]
       add eax,DWORD PTR [rdx+r10*4+0x2c] ;*iadd
                                          ; - LoopUnrolling::hotMethod@14 (line 25)
       add r9d,0x8  ;*iinc
                    ; - LoopUnrolling::hotMethod@16 (line 24)
       cmp r9d,r8d
       jl L0000  ;*if_icmpge
                 ; - LoopUnrolling::hotMethod@7 (line 24)
// handle the remaining array elements
...
```

Loop Unrolling

```
// peels the first loop iteration
sum = array[0]

i = 1
L0000:
      sum += array[i + 0]
      sum += array[i + 1]
      sum += array[i + 2]
      sum += array[i + 3]
      sum += array[i + 4]
      sum += array[i + 5]
      sum += array[i + 7]
      i += 8
cmp i,array.length
jl L0000

// handle the remaining array elements
```

Quizz

- **Q: Would it be a good idea to do manual loop unrolling?**

```
private static int hotMethod(int[] array) {
    int sum = 0;

    for (int i = 0; i < array.length; i += 4) {
        sum += array[i];
        sum += array[i + 1];
        sum += array[i + 2];
        sum += array[i + 3];
    }

    // handle the remaining array elements
    return sum;
}
```

Quizz

- **A: Not really, let the JIT do the optimization by itself for your code …**

```
// peels the first loop iteration
sum = array[0]
sum += array[1]
sum += array[2]
sum += array[3]

i = 4
L0000:
      sum += array[i]
      sum += array[i + 1]
      sum += array[i + 2]
      sum += array[i + 3]
      sum += array[i + 4]
      sum += array[i + 5]
      sum += array[i + 7]
      i += 8
cmp i,array.length
jl L0000

// handle the remaining array elements
```

JIT recognizes it as an unrolled loop and **rerolls** it before **unrolling** it again … ☺

Loop Unrolling

Sum Up

Loop unrolling reduces the number of branches, hence removes some of the overhead of iterating through the entire loop.

Loop unrolling increases the degree of instruction level parallelism and might allow basic block level vectorization.

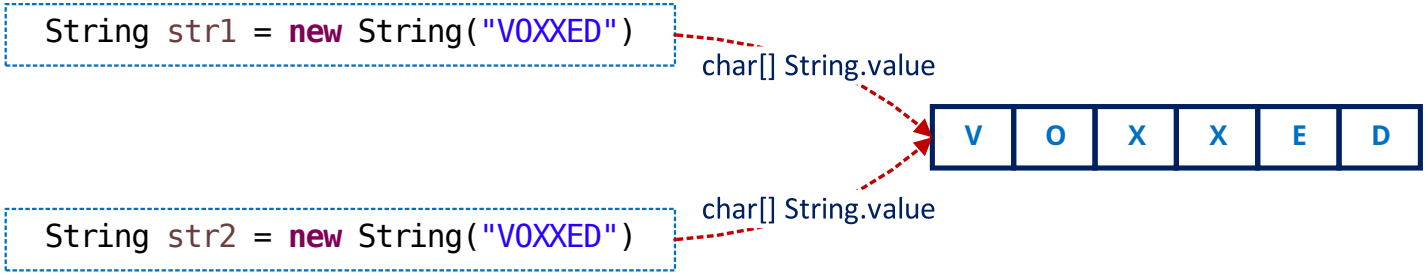
Avoid doing manual unrolling.

String Deduplication

String Deduplication

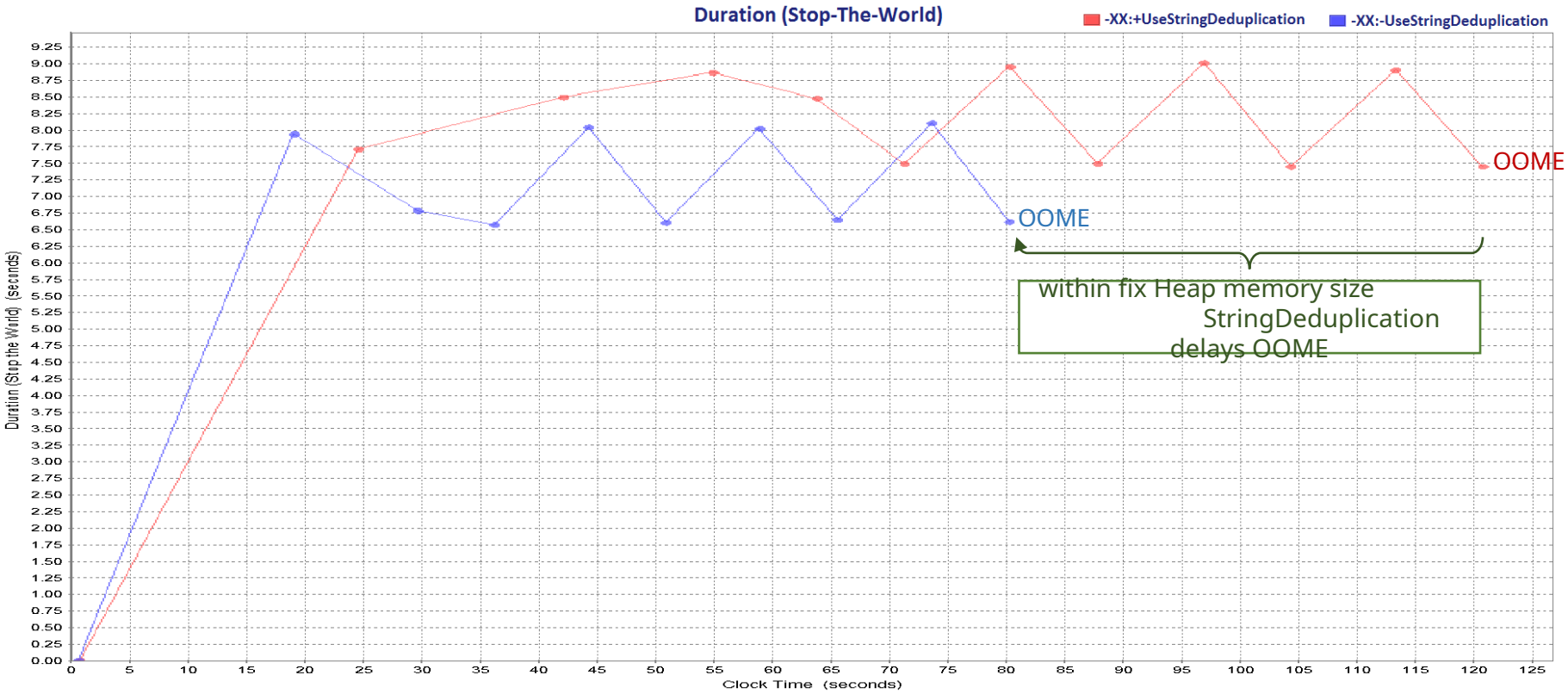
String Deduplication – Strings with equal hashCode() and the same underlying char[] become backed by just one underlying char[]

- underlying char[] no longer referenced are garbage collected
- processes strings which survived few GC cycles -XX:StringDeduplicationAgeThreshold
- available from Java 8 u20 with -XX:+UseG1GC



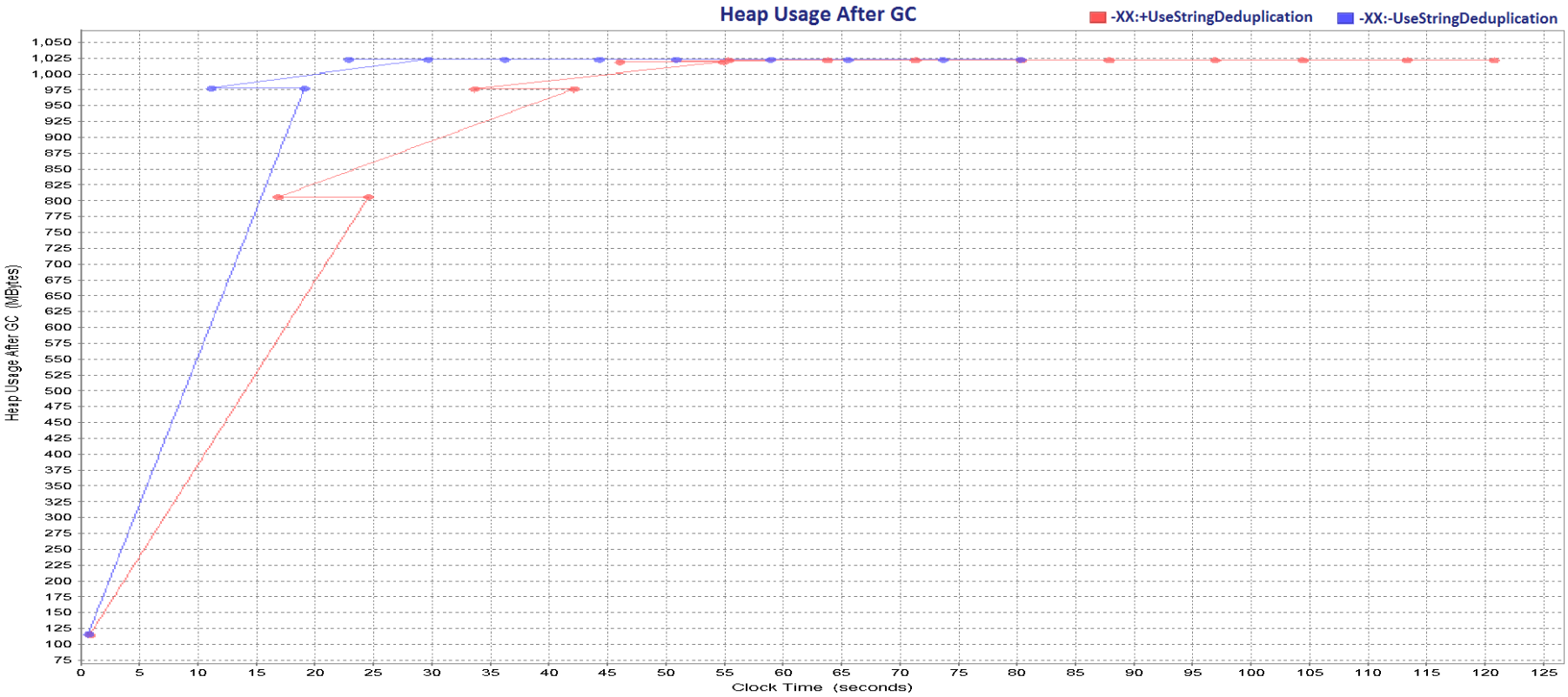
String Deduplication

Measure the impact of creating String objects using -Xmx1024M -XX:+UseG1GC



String Deduplication

Measure the impact of creating String objects using -Xmx1024M -XX:+UseG1GC



String Deduplication

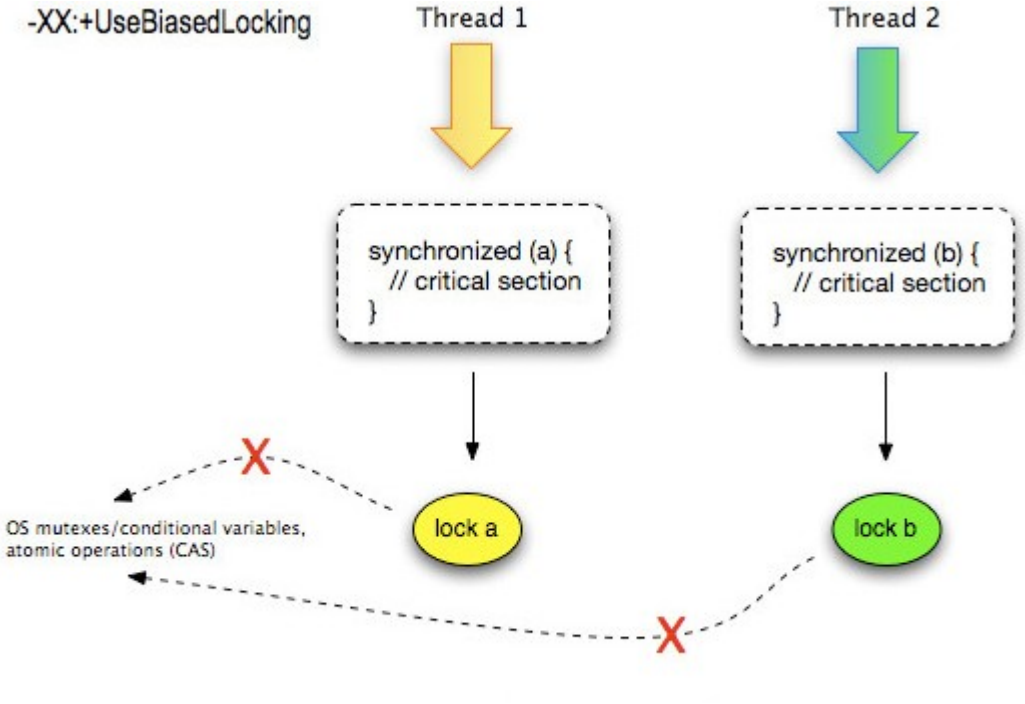
Sum Up

String Deduplication reduces the memory footprint and it is very useful in case of applications with a bounded amount of Heap memory

String Deduplication impacts performance due to time spent in matching the underlying char[] during GC cycles

Biased Locking

Biased Locking



Source: [http://work.tinou.com/2009/06/lock-coarsening-biased-locking-escape-analysis-for-dummies.html]

Biased Locking

- **Code Sample**

```
for (int i = 0; i < NO_OF_THREADS; i++) { // 50 threads
    Thread myThread = new Thread() {

        @Override
        public void run() {

            cyclicBarrier.await();

            synchronized (sharedInstance) {
                sharedInstance.counter++;
            }
        }
    };
    threads[i] = myThread;
}

// start threads
// join threads
```

Biased Locking

$> java -XX:BiasedLockingStartupDelay=0 -XX:+PrintSafepointStatistics -XX:+UnlockDiagnosticVMOptions
-XX:+PrintGCApplicationStoppedTime LoopUnrolling.class

```
Total time for which application threads were stopped: 0.0012160 seconds, Stopping threads took: 0.0011872 seconds
Total time for which application threads were stopped: 0.0000909 seconds, Stopping threads took: 0.0000249 seconds
Total time for which application threads were stopped: 0.0000636 seconds, Stopping threads took: 0.0000273 seconds
Total time for which application threads were stopped: 0.0000774 seconds, Stopping threads took: 0.0000419 seconds
Total time for which application threads were stopped: 0.0000470 seconds, Stopping threads took: 0.0000249 seconds
Total time for which application threads were stopped: 0.0000830 seconds, Stopping threads took: 0.0000320 seconds
...

         vmop                [threads: total initially_running wait_to_block]    [time: spin block sync cleanup vmop] page_trap_count
0.087: EnableBiasedLocking      [      10          0              1    ]         [     0     1     1     0     0    ]  0
0.148: RevokeBias               [      61          0              1    ]         [     0     0     0     0     0    ]  0
...
0.149: BulkRevokeBias           [      43          0              2    ]         [     0     0     0     0     0    ]  0
0.150: RevokeBias               [      25          0              1    ]         [     0     0     0     0     0    ]  0
...
0.151: BulkRevokeBias           [      15          0              2    ]         [     0     0     0     0     0    ]  0
0.152: no vm operation          [      10          0              1    ]         [     0     0     0     0   316    ]  0

Polling page always armed
EnableBiasedLocking                1
RevokeBias                        15
BulkRevokeBias                     2
    5 VM operations coalesced during safepoint
Maximum sync time      1 ms
Maximum vm operation time (except for Exit VM operation)      0 ms
```

Total time for which application threads were stopped: 0.0028275 seconds

Biased Locking

$> java -XX:-UseBiasedLocking -XX:+PrintSafepointStatistics -XX:+UnlockDiagnosticVMOptions -XX:+PrintGCApplicationStoppedTime LoopUnrolling.class

```
         vmop                 [threads: total initially_running wait_to_block]    [time: spin block sync cleanup vmop] page_trap_count
0.315: no vm operation          [      10          0              1    ]          [     0     0     0     0   314    ]  0

Polling page always armed
    0 VM operations coalesced during safepoint
Maximum sync time      0 ms
Maximum vm operation time (except for Exit VM operation)      0 ms
```

Total time for which application threads were stopped: 0 seconds

Biased Locking

Sum Up

Biased locking helps a lot in case of un-contented locks, but if there is any contention it pays the cost of Stop-The-World pauses (e.g. RevokeBias).

In case your application deals with a lot of contention which might be known in advance, may be it is better to disable biased locking.

Inlining & Concurrency Implications

Inlining

It saves calls stack **POP/PUSH** overhead - *naively* ☺

It allows for powerful optimizations:

- if methods do not raise exceptions (consistent control flow) → JIT can move code around
- allows optimizations across method boundaries → optimize caller and callee as a big unit rather than in isolation
- if memory is not changed between method calls → saves number of memory loads

Inlining

- **Code Sample**

```
int sum = 0;
for (int i = 0; i < 20_000; i++) {
    sum += hotMethod();
}
```

```
private double hotMethod() {

    double h1 = rectangle.getHeight();    // return 42.0
    cantInlineMethod();
    double h2 = rectangle.getHeight();
    cantInlineMethod();
    double w1 = rectangle.getWidth();     // return 37.0
    cantInlineMethod();
    double w2 = rectangle.getWidth();

    return ((h1 * w1) + (h2 * w2));
}
```

Inlining

- **Code Sample**

```
private int cantInlineMethod() { return m2(); }
private int m2()  { return m3();  }
private int m3()  { return m4();  }
private int m4()  { return m5();  }
private int m5()  { return m6();  }
private int m6()  { return m7();  }
private int m7()  { return m8();  }
private int m8()  { return m9();  }
private int m9()  { return m10(); }
private int m10() { return m11(); }
private int m11() { return 0;     }
```

```
$> java -XX:+UnlockDiagnosticVMOptions -XX:+PrintFlagsFinal -version | grep InlineLevel
        intx MaxInlineLevel          = 9         {product}
```

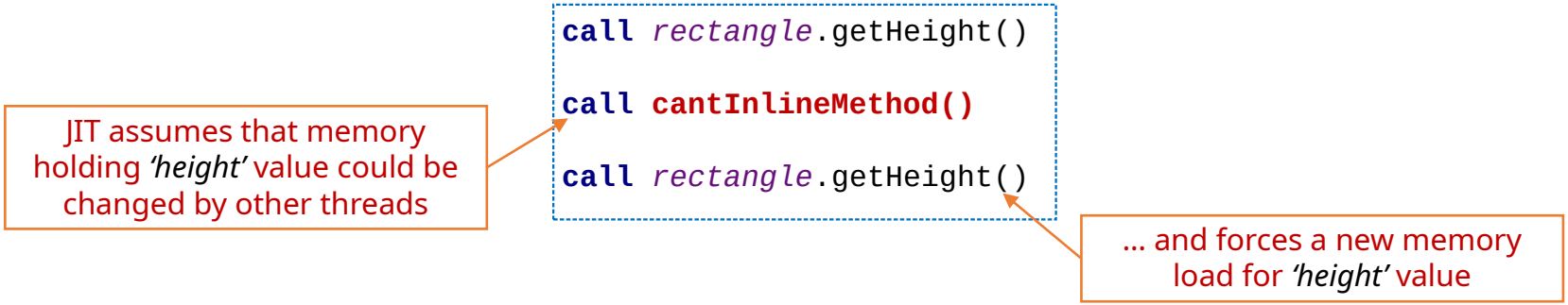
Inlining

$> java -XX:+PrintAssembly -XX:PrintAssemblyOptions=intel -XX:-TieredCompilation LoopUnrolling.class

```
# {method} {0x000000000f650748} 'hotMethod' '()D' in 'SuccessiveMemoryLoads'
#           [sp+0x40]  (sp of caller)
[Entry Point]
...
vmovsd xmm0,QWORD PTR [r12+r11*8+0x10]  ;*getfield height
...
call 0x00000000025f6620  ; OopMap{rbp=Oop off=44}
                         ; - SuccessiveMemoryLoads::cantInlineMethod@0 (line 32)
...
vmovsd xmm0,QWORD PTR [r12+r10*8+0x10]  ;*getfield height
...
call 0x00000000025f6620  ; OopMap{rbp=Oop off=68}
                         ; - SuccessiveMemoryLoads::cantInlineMethod@0 (line 32)


...
vmovsd xmm0,QWORD PTR [r12+r11*8+0x18]  ;*getfield width
...
call 0x00000000025f6620  ; OopMap{rbp=Oop off=92}
                         ; - SuccessiveMemoryLoads::cantInlineMethod@0 (line 32)
...
vmovsd xmm0,QWORD PTR [r12+r10*8+0x18]  ;*getfield width
...
```

Inlining


call rectangle.getHeight()
call cantInlineMethod()
call rectangle.getHeight()
JIT assumes that memory holding 'height' value could be changed by other threads
... and forces a new memory load for 'height' value

Inlining

$> java -XX:+PrintAssembly -XX:PrintAssemblyOptions=intel ***-XX:MaxInlineLevel=12*** *-XX:-TieredCompilation LoopUnrolling.class*

```
# {method} {0x000000000f8a0748} 'hotMethod' '()D' in 'SuccessiveMemoryLoads'
#           [sp+0x20]  (sp of caller)
[Entry Point]
...
// xmm0 = height x width
vmovsd xmm0,QWORD PTR [r12+r11*8+0x18]      ; implicit exception dispatches to 0x000000000282a9f8
vmulsd xmm0,xmm0,QWORD PTR [r12+r11*8+0x10] ;*dmul
                                            ; - SuccessiveMemoryLoads::hotMethod@45 (line 36)
// xmm0 = xmm0 x xmm0
vaddsd xmm0,xmm0,xmm0                       ;*dadd
                                            ; - SuccessiveMemoryLoads::hotMethod@50 (line 36)
...
```

Because cantInlineMethod() was now inlined, it reduced the memory loads in case of *height* and *width.*

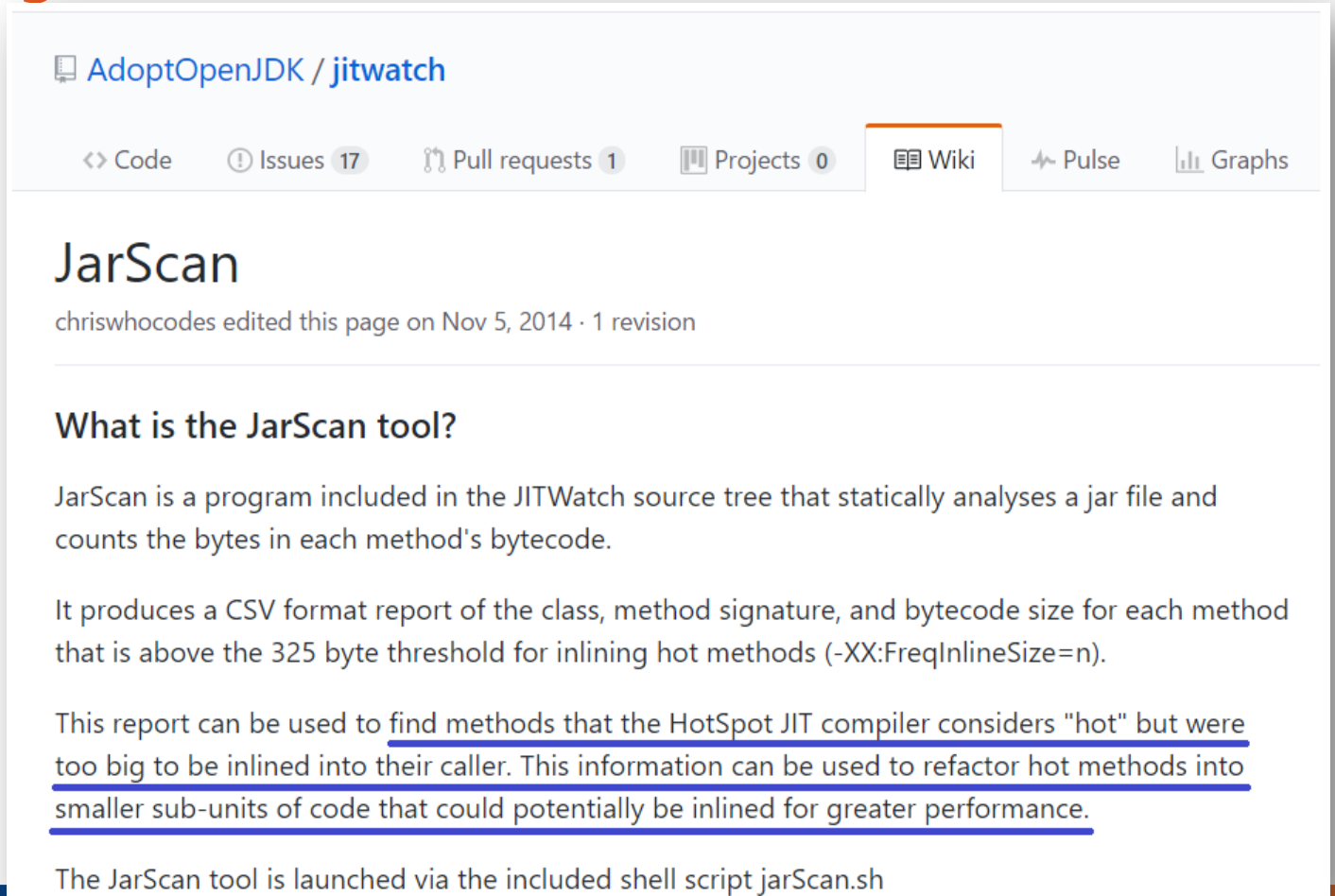
Inlining

Sum Up

Inlining open doors towards other JIT optimizations!

-XX:MaxInlineSize	Default is 35 bytes
-XX:MinInliningThreshold	Default is 250
-XX:FreqInlineSize	Default is 325 bytes
-XX:MaxInlineLevel	Default is 9
-XX:MaxRecursiveInlineLevel	Default is 1

Inlining



AdoptOpenJDK / jitwatch

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JarScan

chriswhocodes edited this page on Nov 5, 2014 · 1 revision

What is the JarScan tool?

JarScan is a program included in the JITWatch source tree that statically analyses a jar file and counts the bytes in each method's bytecode.

It produces a CSV format report of the class, method signature, and bytecode size for each method that is above the 325 byte threshold for inlining hot methods (-XX:FreqInlineSize=n).

This report can be used to find methods that the HotSpot JIT compiler considers "hot" but were too big to be inlined into their caller. This information can be used to refactor hot methods into smaller sub-units of code that could potentially be inlined for greater performance.

The JarScan tool is launched via the included shell script jarScan.sh

Methods that do not JIT

Huge Methods

- **Code Sample**

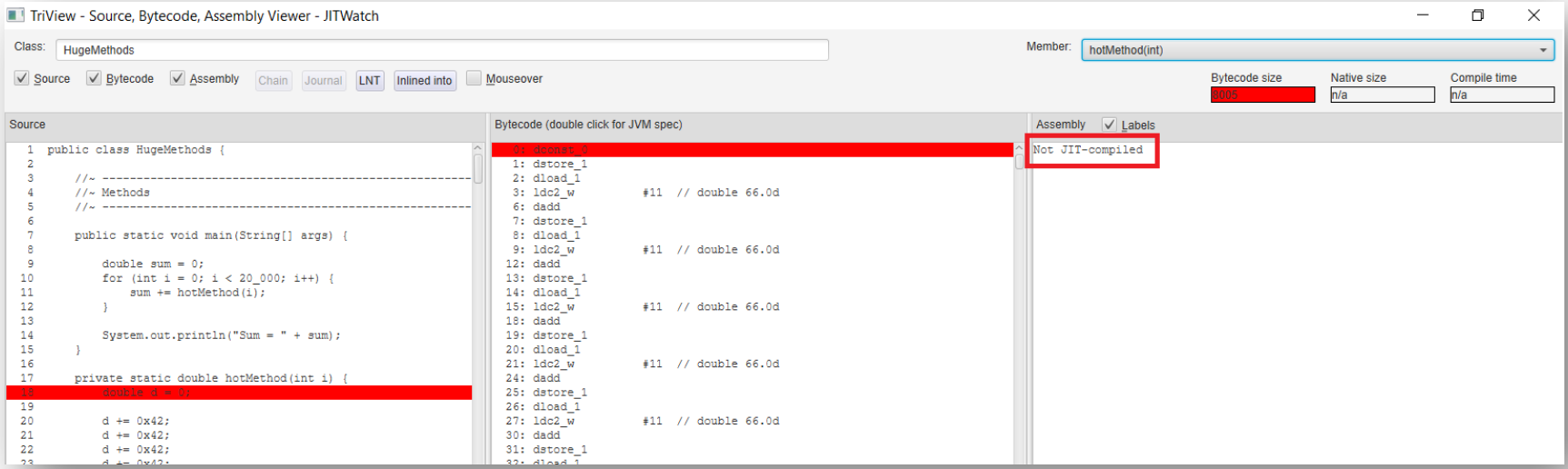
```
private double hotMethod(int i) {
    double d = 0;

    d += 0x42;  // Line 4
    d += 0x42;  // Line 5
    d += 0x42;  // Line 6
    ...
    d += 0x42;  // Line 1335
    d += 0x42;  // Line 1336

    return d + i;
}
```

Method size is 8005 bytes

Huge Methods

$> java -XX:+PrintCompilation -XX:-TieredCompilation HugeMethods.class

```
293    12 %     com.luxoft.jpt.conference.HugeMethods::main @ 4 (50 bytes)
352    12 %     com.luxoft.jpt.conference.HugeMethods::main @ -2 (50 bytes)   made not entrant
```




```
$> java -XX:+UnlockDiagnosticVMOptions -XX:+PrintFlagsFinal -version | grep DontCompile
        bool DontCompileHugeMethods                 = true     {product}
```

Huge Methods

$> java -XX:+PrintCompilation -XX:-TieredCompilation -XX:-DontCompileHugeMethods HugeMethods.class

```
234   12       com.luxoft.jpt.conference.HugeMethods::hotMethod (8005 bytes)
265   13 %     com.luxoft.jpt.conference.HugeMethods::main @ 4 (50 bytes)
```

JDK-6453531 : Severe performance regression on long methods for server VM

Details

Type: Bug	**Submit Date:** 2006-07-27
Status: Closed	**Updated Date:** 2011-02-23
Project Name: JDK	**Resolved Date:** 2006-07-27
Component: hotspot	**OS:** linux
Sub-Component: compiler	**CPU:** x86

Comments

EVALUATION

Compilation of very large methods (>8000 bytes) is not well-tested and expensive. The VM is tuned for compiling for the common programming paradigm, which includes generally small methods.

The user can work around the issue with the VM flag -XX:-DontCompileHugeMethods.

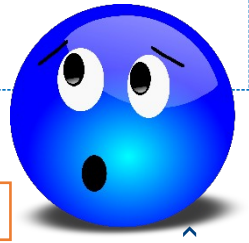
WORK AROUND

Use the VM flag -XX:-DontCompileHugeMethods

Too Many Arguments

Code Sample

```
private double hotMethod(
        //J+
        double d00, double d01, double d02, double d03, double d04, double d05, double d06, double d07, double d08, double d09,
        double d10, double d11, double d12, double d13, double d14, double d15, double d16, double d17, double d18, double d19,
        double d20, double d21, double d22, double d23, double d24, double d25, double d26, double d27, double d28, double d29,
        double d30, double d31, double d32, double d33, double d34, double d35, double d36, double d37, double d38, double d39,
        double d40, double d41, double d42, double d43, double d44, double d45, double d46, double d47, double d48, double d49,
        double d50, double d51, double d52, double d53, double d54, double d55, double d56, double d57, double d58, double d59,
        double d60, double d61, double d62, double d63, double d64, double d65, double d66, double d67, double d68, double d69
        //J-
) {

    return
        //J-
        d00 + d01 + d02 + d03 + d04 + d05 + d06 + d07 + d08 + d09 + d10 + d11 + d12 + d13 + d14 + d15 + d16 + d17 + d18 + d19 +
        d20 + d21 + d22 + d23 + d24 + d25 + d26 + d27 + d28 + d29 + d30 + d31 + d32 + d33 + d34 + d35 + d36 + d37 + d38 + d39 +
        d40 + d41 + d42 + d43 + d44 + d45 + d46 + d47 + d48 + d49 + d50 + d51 + d52 + d53 + d54 + d55 + d56 + d57 + d58 + d59 +
        d60 + d61 + d62 + d63 + d64 + d65 + d66 + d67 + d68 + d69;
        //J+
}
```

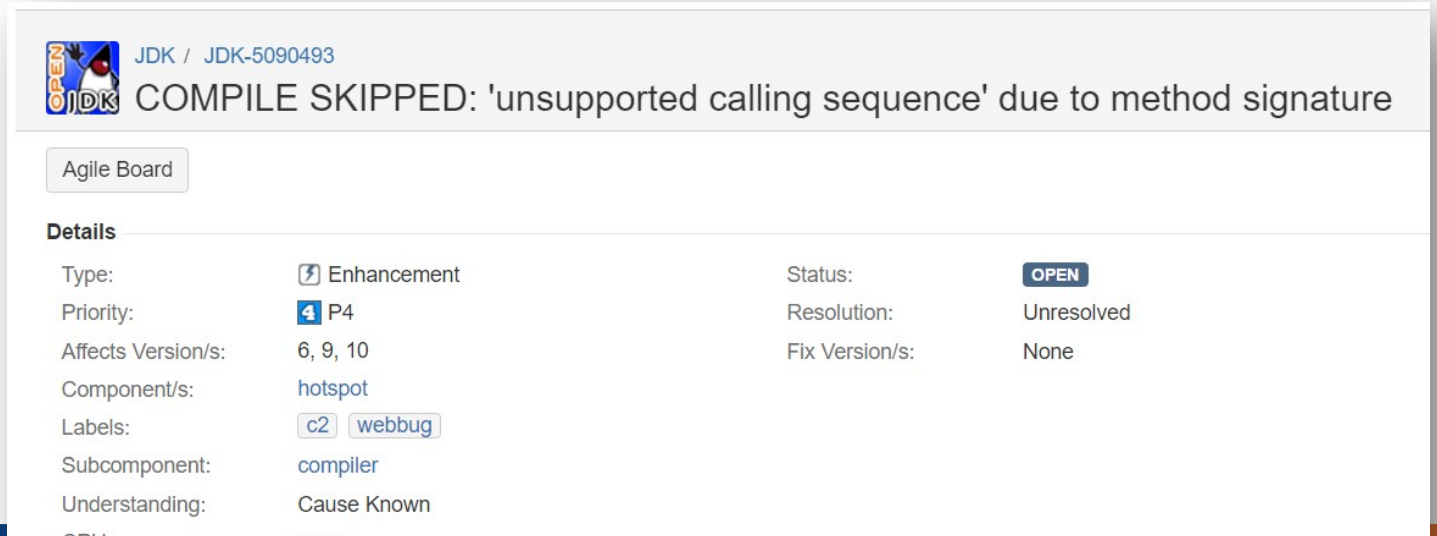


Method has 70 doubles arguments

Too Many Arguments

$> java -XX:+PrintCompilation -XX:-TieredCompilation TooManyArgs.class

```
155    12       com.luxoft.jpt.conference.TooManyArgs::hotMethod (208 bytes)
156    12       com.luxoft.jpt.conference.TooManyArgs::hotMethod (208 bytes)
                        COMPILE SKIPPED: unsupported incoming calling sequence (not retryable)
157    13 %     com.luxoft.jpt.conference.TooManyArgs::main @ 2 (226 bytes)
158    13 %     com.luxoft.jpt.conference.TooManyArgs::main @ 2 (226 bytes)
                        COMPILE SKIPPED: unsupported calling sequence (not retryable)
```



In Closing …

“It doesn't make a difference how beautiful your guess is. It doesn't make a difference how smart you are, who made the guess, or what his name is.

If it disagrees with experiment, it's wrong.”

Richard Feynman

The Feynman Lectures on Physics: The New Millennium Edition

In Closing …

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Richard Feynman

The Feynman Lectures on Physics: The New Millennium Edition

Always keep on experiencing new things !

THANK YOU

Ionuţ Baloşin

Software Architect
ibalosin@luxoft.com
@ibalosin

LUXOFT

LXFT LISTED NYSE

Appendix

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Java performance techniques. The cost of HotSpot runtime optimizations.

Ionuţ Baloşin

Software Architect
ibalosin@luxoft.com
@ibalosin

Just In Time Compiler - HotSpot

Tiered Compilation Flows